Pre-analysis plan

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H1. ***Having daughters leads to stronger support for gender equality (as measured by attitudes).***

Dep. Variable

- plh0298_v2 ("Child below age 6 suffers when mother works")
- only measured in 2018
- agreement measured on 7-point Likert scale
- recoded so that higher numbers mean higher disagreement, i.e. greater support for gender equality

Main Independent Variable

- Having a daughter dummy (1=yes, 0=no)
- First step was to construct the number of daughters by respondent and survey year using kidsex, kidgeb, and syear variables
- Second step was to construct a dummy variable for having a daughter at point t (d_daughter)
- (Important also to construct number of sons, needed to construct number of children)

Control variables

- Sex of respondent (male)
- 10-year birth cohorts (constructed out of gebjahr)
- Years of schooling (pgbilzeit)
- Left wing political preferences (leftwing): see construction below (H2)
- Dummy for married
- Dummy for born in Germany
- Number of children (see above, DO NOT use sumkids!!)
- Ln household income (lnhhinc) from hghinc
- Dummy for living in East Germany (eastgerm)
- For all variables: missing values were coded from minus values to "."

Inclusion/ Exclusion

- All respondents to the core SOEP that participated in the 2018 wave
- Excluded are only those that had missing information in any of the variables included in the model

Exact analysis/ Identification

- Instrumental variables estimation using cross-sectional wave of 2018

- Instrument: sex of first child (firstgirl)
    - Established instrument in the literature, see e.g. Washington (2008) in AER
- Robust standard errors
- STATA code: ivreg2 plh0298_v2 male birthc* pgbilzeit leftwing mar germborn numchild lnhhinc eastgerm (d_daughter = firstgirl), r first

Interpretation of results

- Coefficient, Z-statistic, and p-value of d_daughter variable
- If positive and significant (p-value<0.05), H1 is supported
- Instrument relevance check: F-statistic of excluded instrument (should be greater than 10)
- Exclusion restriction cannot directly be tested (via Sargan or Hansen), but theoretically it can be argued that the sex of the first child is randomly assigned (see Washington, 2008)

H2. ***Having daughters leads to more left-leaning political preferences.***

Dep. Variable

- Dummy for having left-leaning political preferences (leftwing)
- Constructed out of plh0012_h
- Parties or coalitions considered left-leaning include: SPD, Linke, Greens, Pirates

Independent Variable

- d_daughter (same as above) lagged by 3 years

Control variables

- same as above (time-invariant ones dropped automatically; within regression); plus
    - partypreference (yes/no): constructed out of plh0011_h
    - employment of respondent (dummies for pgemplst)
    - survey year dummies

Inclusion/ Exclusion

- Largest available sample from 2000 to 2020
- Excluded are only those observations that had missing information in any of the variables included in the model

Exact analysis/ Identification

- Fixed effects estimation of panel data from 2000 to 2020
- Independent variable (d_daughter) lagged by 3 years

- Correction for clustering of standard errors
- STATA code:
    - xtset pid syear
    - xtreg leftwing partypreference l3.d_daughter pgbilzeit pgemplstd2-pgemplstd6 eastgerm i.syear, fe vce(cluster pid)

Interpretation of results

- Coefficient, t-statistic, and p-value of lagged d_daughter variable
- If positive and significant (p-value<0.05), H2 is supported

H3. ***Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).***

Dep. Variable

- Average weekly working hours for pay or profit (pgtatzeit)

Independent Variable

- d_daughter (same as above) lagged by 3 years

Control variables

- same as H1 model (time-invariant ones dropped automatically; within regression); plus
    - number of children age 0-5
    - survey year dummies

Inclusion/ Exclusion

- Largest available sample from 2000 to 2020
- Sample restrictions: only married women in working age (25 to 64)
- Intuition: if women work more it might suggest greater gender equality among married couples (even though it might also indicate less investment in time spent with daughters compared to sons…)
- Excluded are only those observations that had missing information in any of the variables included in the model

Exact analysis/ Identification

- Fixed effects estimation of panel data from 2000 to 2020
- Independent variable (d_daughter) lagged by 3 years
- Correction for clustering of standard errors
- STATA code:

    - xtset pid syear
    - xtreg workhours l3.d_daughter pgbilzeit mar numchild* eastgerm syeard* if male==0 & mar==1 & age>=25 & age<65, fe vce(cluster pid)

Interpretation of results

- Coefficient, t-statistic, and p-value of lagged d_daughter variable
- If positive and significant (p-value<0.05), H3 is supported

H4. ***Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).***

- Same model as H1
- Newly added to the model: interaction term of d_daughter and male (d_daughter_male)
- Same estimation strategy as H1
- Instrumented variables: d_daughter, d_daughter_male
- Instruments: firstgirl, firstgirl_male
- STATA code:
    - gen d_daughter_male = d_daughter * male
    - gen firstgirl_male = firstgirl * male
    - ivreg2 plh0298_v2 male birthc* pgbilzeit leftwing mar germborn numchild lnhhinc eastgerm (d_daughter d_daughter_male = firstgirl firstgirl_male), r first